

ANDREAS LOEHR

ML Engineer & Part-time PhD Student in Reinforcement Learning

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📅 14 October 1997

📍 Frankfurt, Germany

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Mathematics-driven Software Engineer with focus on Machine Learning and Reinforcement Learning. Experience integrating ML solutions into production systems, building distributed systems, and conducting applied research. Passionate about understanding systems at depth.

EXPERIENCE

Software Engineer - Machine Learning

SimPlan AG

📅 Nov 2024 – Ongoing

📍 Hanau, Germany

Integrated custom Machine Learning solutions into value stream modeling (VSM) software SimVSM.

- Value Stream Digitization: Iterated on ML backend for digitizing analog value streams (c.f. Working Student section). Implemented image preprocessing pipeline to handle large panorama images, refactored backend to use task queues for robustness, and deployed via Docker Compose with Sentry monitoring.
- Value Stream Assistant: Built and integrated agentic AI assistant with voice input (Web Speech API), evaluation pipeline. Iterative prototyping for assistant improvements.

Working Student - Software Engineering

SimPlan AG

📅 Nov 2022 – Nov 2024

📍 Hanau, Germany

Implemented ML pipeline for value stream digitization in VSM software.

- Built three-step computer vision pipeline: object detection, OCR text detection, and edge detection for digitizing analog value streams (edge detection just integrated, mostly done by our university partner).
- Developed custom backend API, conducted data collection and annotation, trained object and text detection models.

Student Worker - Pandemic Modeling Tool

Goethe-University Frankfurt

📅 Feb 2022 – May 2022

📍 Frankfurt, Germany

Translated discrete-time epidemic/pandemic modeling tool from Mathematica to Python.

Working Student - Risk Management

Deloitte Risk Advisory

📅 June 2019 – May 2021

📍 Frankfurt, Germany

Developed text mining and data analysis tools for financial risk assessment.

- Built MS Access/VBA frontend for SQL Server backend (NII stress testing tool): document upload, analysis, simulation validation, and automated report generation with unit and integration tests.

EDUCATION

PhD Reinforcement Learning

Karlsruher Institut für Technologie (KIT)

📅 Aug 2025 – Ongoing

- Part-time PhD; drafting research proposal; active research group engagement

M.Sc. Mathematics

Goethe-University Frankfurt

📅 April 2022 – Nov 2024

Thesis: *Policy Evaluation in Distributional Reinforcement Learning* (Code on GitHub)

- Grade: 1.3
- Specialization: Probability Theory (Stochastic Analysis, Advanced Stochastics)
- Electives: Numerical Mathematics, Functional Analysis
- Minor: Computer Science (ML, Distributed Systems)

B.Sc. Mathematics

Goethe-University Frankfurt

📅 April 2019 – Aug 2022

Thesis: *Learning Using Statistical Invariants* (Code on GitHub)

- Grade: 2.3
- Specialization: Probability Theory; Minor: Computer Science

B.Sc. Business Administration

University of Mannheim

📅 Aug 2016 – July 2019

- Grade: 2.2
- Business fundamentals, finance, and management; concurrent with math studies

Semester Abroad

Singapore Management University

📅 Aug 2018 – Dec 2018

- Named Entity Recognition prototype using Transformers; improved duplicate detection via regex and similarity measures; image preprocessing and OCR.

Internship - Risk Management

Deloitte Risk Advisory

📅 January 2019 – March 2019 📍 Frankfurt, Germany

Image preprocessing and OCR enhancement for financial documents.

- Developed image preprocessing pipeline using OpenCV to improve OCR accuracy on document scans.
- Applied techniques: noise reduction, binarization, deskewing; integrated into document analysis workflow.

PROJECTS

Real-time Embedded Sensing on ESP32 via ESP-IDF

Personal Project

📅 2026 📍 GitHub

Low-level embedded C++ application for sensor data collection on microcontroller with FreeRTOS-based architecture.

- Task-based concurrency model (WifiConnectTask, Sht3xTask, UpdateTask, TimeSyncTask) with queue-based sensor data flow and event group synchronization.
- Direct microcontroller hardware: I2C protocol handling, SHT31 sensor integration, NVS-based storage for credentials.
- HTTP client for pushing real-time telemetry; designed for deterministic timing on embedded platform.

ONNX Model Inference & Serving in C++

Personal Project

📅 2025 📍 GitHub

Prototypical C++ inference server with AMPQ Message Routing for deploying and serving machine learning models.

- Model inference: Expose YOLOv11 object detection models via onnxruntime C++ API and AMQP message routing;
- OnnxInferLib abstraction layer for ONNX runtime inference with image preprocessing and post-processing pipelines.
- Testing: C++ GTest-based unit tests; Python client for results validation.

- Focus: Quantitative Finance, Statistics, Programming

SKILLS

Python C/C++ Embedded Systems

Microcontrollers

Machine Learning

Model Inference

ONNX

PyTorch

ESP32

FreeRTOS

ESP-IDF

CMake

FastAPI

Docker

Linux

Git

REST APIs

CI/CD

Real-time Systems

Computer Vision

LANGUAGES

German - Native

English - C1

Spanish - B1

Russian - A1

PUBLICATIONS

References

- Micha Selak et al. (2026). *Agentic Insight Generation in VSM Simulations*. arXiv: 2604.12421 [cs.CL]. URL: <https://arxiv.org/abs/2604.12421>.
- – (Dec. 2025). “LLM Assisted Value Stream Mapping”. In: *Proceedings of the 2025 Winter Simulation Conference*. WSC, pp. 2015–2026. DOI: 10.1109/WSC68292.2025.11338928.

INTERESTS

Embedded Systems, Real-time AI, Microcontrollers, Language Learning, Fitness & Nutrition, Cooking & Fermentation